

FREE

SSRIs are Mitochondrial Drugs

Installment six in our series on understanding the truth about SSRIs.



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JUL 13, 2025



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Our [series](#) has so far focused on serotonin and melatonin but now turns to SSRIs themselves. These not only interact with the ability of serotonin and melatonin to support mitochondrial function in both good and bad ways, but are *themselves* drugs that act on mitochondria both by activating receptors that have little to do with serotonin inside cells and by traveling to the mitochondria and having variable effects that are mostly toxic.

SSRIs are called “selective” serotonin reuptake inhibitors because they inhibit serotonin transporters that bring serotonin into cells but do not meaningfully inhibit the transporters of other neurotransmitters like dopamine.

This name is grossly at odds with the name of serotonin itself, because the word “reuptake” implies that this is primarily relevant to synapses in the nervous system. Synapses are where one neuron meets another. The first neuron releases serotonin into the synapse where it can act on the second neuron. To ensure this activity does not last longer than it should, “reuptake” moves serotonin back into the first neuron so that it is no longer stimulating the second neuron. That is, it is taken *back* or taken up *again* in another event within a repeated cycle, so we call it *re*-uptake.

But serotonin is named after its presence in the *blood* and its action on *smooth muscle cells* where it stimulates contraction by acting directly on the serotonin receptors of the muscle cells. When these muscle cells contract, our blood vessels narrow, known as vasoconstriction.

Serotonin travels into the cells that it acts on through the serotonin transporter, which is “uptake,” but not “reuptake.”

SSRIs should be called *uptake* inhibitors, not *reuptake* inhibitors, because they inhibit the uptake of serotonin *everywhere*.

As pointed out in [installment three](#), these transporters [are found](#) as abundantly in the gastrointestinal tract as in the brain, and are found at competitive abundance in the respiratory system and in male and female reproductive tissues. They are found at meaningful abundance in the liver and gall bladder, urinary system, pancreas, and urinary tissues, and they are not absent from any tissue.

Neurons are minor expressers of the transporter. The [cell types](#) with the most expression are the enterocytes of the gut and the trophoblasts of the placenta. There is considerable expression by all sorts of glandular cells, skin cells, lung cells, sperm cells, heart muscle cells, smooth muscle cells, skeletal muscle cells, fat cells,

connective tissue cells, immune cells, and blood cells that all rival the expression in neurons, many of which exceed the expression in neurons.

SSRIs inhibit the uptake of serotonin into all of these cells, where the term “reuptake” has no application.

That these drugs are “selective” only applies to the *relative reuptake of different neurotransmitters*.

That is, they are not “selective” toward serotonin reuptake in the sense that they inhibit this more than they do completely unrelated things.

No, they are only “selective” toward this in that they don’t also inhibit the reuptake of other neurotransmitters like dopamine.

Mainstream medicine and pharmacology primarily view the significance of “off-target” effects of SSRIs as increasing extracellular serotonin in the gut where it activates 5-HT₃ and 5-HT₄ receptors that can increase gastrointestinal upset, nausea, and vomiting, and negative feedback inhibition of serotonin release causing a two-week lag in the antidepressant effect.

Mainstream medicine and pharmacology primarily view the differences between SSRIs as a matter of their [different strength](#) in inhibiting serotonin transporters, which can cause differences in the dose required or the speed of therapeutic action.

However, SSRIs travel into the cell such that the free concentration inside cells will always equilibrate with the free concentration outside cells. Some travel into cells much more than others, which makes the excess accumulate in cell membranes. SSRIs can be classed as strong, moderate, or weak activators of the **sigma-1 receptor**, which has powerful impacts on psychiatric responses that are mediated in large part by its direct impacts on mitochondrial function. From there, SSRIs are then free to have their own direct impacts on mitochondria.

The Serotonin System Is Hard to Hack With SSRIs

The mainstream pharmacological paradigm led to various efforts to try to better hack the serotonin system that never quite solved the problems of eliminating nausea or the initial delay in therapeutic activity.

[From the 1990s](#), it was a holy grail of SSRI development to find a way to inhibit the serotonin transporter while directly inhibiting the 5-HT_{1A} receptors on the presynaptic neuron while stimulating the 5-HT_{1A} receptors on the postsynaptic neuron. This would increase synaptic serotonin and abolish the negative feedback against serotonin release on the presynaptic neuron, all without antagonizing the impact of serotonin in regulating aggression, anxiety, addiction, appetite, memory, mood, pain perception, sleep, and body heat all mediated by the postsynaptic 5-HT_{1A} receptors. In theory, this would abolish the initial delay in therapeutic efficacy without causing psychiatric harm.

This, of course, was a laughable pipe dream. The main tool in Pharma’s kit was to play around with the ratio of serotonin transport inhibition to 5-HT_{1A} inhibition or

activation, but this does nothing to distinguish between presynaptic and postsynaptic 5-HT_{1A} receptors.

This effort spawned vilazodone (Viibryd), which is considered a serotonin modulator but not an SSRI. It inhibits serotonin transport and acts as a partial 5-HT_{1A} activator, and the main effect was that it is much worse than SSRIs in producing nausea and so has to be titrated up to the effective dose much more carefully.

It also spawned vortioxetine (Trintillex). This also is not an SSRI. It inhibits serotonin transport, inhibits three of the fifteen serotonin receptors, partially activates one serotonin receptor, fully activates another, and increases signaling activity of dopamine, norepinephrine, acetylcholine, glutamate, and GABA. It has a lower rate of side effects than SSRIs, especially with regard to sexual dysfunction and sleep disruption, but it still causes transient nausea in over 20% of users, a withdrawal rate of 8% over the course of a year due to adverse effects, and a long-term incidence of sexual dysfunction in over 1% of users.

[Litoxetine](#) was supposed to be an SSRI that inhibits 5-HT₃ receptors and for that reason has anti-nausea instead of pro-nausea effects. However, it was withdrawn from the approval process in the 1990s for undisclosed reasons and is [now being reconsidered](#) in France, Poland, and the UK as a drug for urinary incontinence. We do not know why it was never fully pursued for depression, but perhaps this is related to the role of 5-HT₃ receptors in the nervous system where they regulate cognitive function, emotions, appetite, and pain perception, or their occurrence in the mitochondrial membrane where they preserve respiratory chain function during hypoxia.

Another strategy was to use SSRIs alongside pindolol, which inhibits beta-adrenergic receptors as well as 5-HT_{1A} receptors. A handful of trials led to [inconclusive results](#). In general it appeared to not be effective, but a small study suggested this might be because the doses used were not high enough. People complained of tiredness, nausea, vomiting, itching, postural hypotension, sweating, dry mouth, and mild transient dizziness, but it did not seem to change the safety profile compared to SSRIs alone.

What these largely failed efforts to be more pharmacologically precise betray is that Pharma cannot be precise. Your *natural physiology* is incredibly precise, but Pharma works at the inferior level of biochemistry and molecular biology. It can make a drug interact with a receptor, and it can try to add some specificity to the destination of the drug by varying the mode of administration, but it can't make it travel to one particular part of the brain and act on one receptor on one side of a synapse and do the complete opposite on the other side of the synapse.

By contrast, your body makes fifteen different serotonin receptors and has them combine with each other and with many other types of receptors in ways science has barely begun to understand to diversify their functions so that it can, in its inborn wisdom, make serotonin do different things in different places at different times in different contexts. When you give your body what it needs to nourish itself, you work *with* this complex physiology instead of trying to control it using tools that are wholly unfit for the job.

If you want to “hack” this physiology you should be doing it with different holistic stimuli, such as exercise, altitude, hypoxia, hyperoxia, eating, fasting, breathwork, and so on. These stimulate your body to use its own complex physiology to meet the challenge provided by the stimulus. It allows our relatively ignorant minds to focus on the simplicity of the stimulus while our incomprehensibly intelligent bodies respond with their own incredible complexity. We will cover these alternatives at the end of the series.

First, we must sort out the powerful and consuming mitochondrial fire that SSRI prescribers are ignorantly playing with.

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SSRI Brand and Generic Names

In this article I will use the generic names of the SSRIs since these are used in the literature. However, you may be mostly familiar with the brand names. Here is a quick guide to their interconversions:

Prozac — fluoxetine

Celexa (citalopram), a 50/50 mix of escitalopram (also sold separately as Lexapro) and its less effective mirror image isomer

Zoloft — sertraline

Luvox — fluvoxamine

Paxil — paroxetine

SSRIs Massively Deplete Circulating Serotonin

We should expect SSRIs in principle to prevent serotonin from entering cells, which should lower intracellular serotonin and thereby inhibit the beneficial impact of serotonin and melatonin on mitochondrial function during hypoxia.

This should also deplete circulating serotonin.

Indeed, a [human cross-sectional study](#) showed that 64 long-term users of citalopram, escitalopram, fluoxetine, fluvoxamine, paroxetine or sertraline all pooled together had 14-fold lower platelet serotonin than 64 matched controls. However, the different SSRIs were not reported separately. The scatterplot of platelet serotonin shows one individual with a serotonin level comparable to controls, and the only SSRI represented by a single person is fluvoxamine, raising the possibility that fluvoxamine has a divergent effect.

A two-week [double-blind study](#) of paroxetine suggested this effect is achieved within 18 days and is driven largely by serotonin being excreted in the urine instead of accumulating in platelets. However, this effect on *circulating* serotonin might not be very relevant to the tissues that synthesize serotonin locally.

The working model of whole-body serotonin metabolism is that serotonin is primarily made in the gut, is stored in platelets, and is metabolized to 5-hydroxyindoleacetic acid (5-HIAA) in the liver and lungs. Meanwhile, brain serotonin is synthesized in the brainstem and has its metabolism compartmentalized in the brain. 5-HIAA from the brain, liver, lungs, or anywhere else then leaves the body in the urine.

SSRIs appear to be draining serotonin from the blood into the urine instead of allowing circulating serotonin to be stored in platelets.

The lungs synthesize serotonin themselves, but also pick it up from circulating platelets and perhaps the neuroepithelial bodies that sense hypoxia could receive some traveling up the vagus nerve from the gut. Many other tissues outside the brain, however, probably get exposed to serotonin primarily when systemic hypoxia causes its release from platelets. This systemic hypoxia response should therefore be massively impaired in SSRI users.

There seems to be no exploration of this possibility in the literature, but it is notable that airplane travel is a form of hypoxic stress, and SSRI use in pilots [has been implicated](#) in airplane crashes, though with no ability to identify them as having caused the crashes.

SSRIs Have Different Effects on Intracellular Serotonin

It would seem that SSRIs should lower intracellular serotonin in every cell type, since they inhibit the import of serotonin into the cell.

However, this could hypothetically lead to a release of negative feedback against serotonin synthesis within that cell that then compensates for this.

Further, SSRIs are *not selective* to inhibition of serotonin transport when compared to *other unrelated effects inside the cell*. So SSRIs could be independently increasing serotonin synthesis once inside the cell.

The studies we have on how SSRIs impact intracellular serotonin are weak and inadequate, because they are limited to relatively short animal studies where individual SSRIs have been assessed for very different lengths of time in the different studies.

What these studies suggest, though, is that **citalopram and paroxetine may lower serotonin synthesis, while sertraline may raise it.**

[21-day citalopram treatment](#) of rats dramatically lowered total brain serotonin.

In [mouse forebrain](#), citalopram decreased serotonin synthesis substantially within two days; this effect remained strong through 28 days, but was mitigated by decreased serotonin oxidation, causing total serotonin levels to only be slightly and not statistically significantly lower.

Rats treated for three days [with paroxetine](#) had brain serotonin synthesis nearly cut in half.

[Sertraline treatment](#) of leukemia cells in vitro lowers intracellular serotonin over the course of 48 hours due to uptake inhibition, but raises intracellular serotonin above baseline by 96 hours due to increased expression of tryptophan hydroxylase, the first enzyme involved in serotonin synthesis. Two-week treatment of rats with sertraline also raises the expression of tryptophan hydroxylase in the raphe nucleus of the brainstem as well. Fluoxetine also activated the enzyme in this experiment but was not studied in as much detail.

We will soon see that the general pattern emerging from this weak and inadequate spattering of animal research actually lines up perfectly with the relative strength with which these SSRIs activate the sigma-1 receptor.

Some SSRIs Inhibit Some Serotonin Receptors

Fluoxetine binds to 5-HT_{2C} receptors, which various studies have suggested [inhibits](#) or [activates](#) them, while citalopram does so much more weakly and paroxetine has no effect.

Fluoxetine also [inhibits](#) 5-HT_{2A} receptors, which could hypothetically antagonize their promotion of mitochondrial biogenesis.

In guinea pigs, both paroxetine and fluoxetine inhibit the ability of [5-HT₃ and 5-HT₄ receptors](#) in the gut to stimulate intestinal contractions, but fluoxetine is far more potent; neither have much inhibitory power toward 5-HT₄ receptors in the brain but both inhibit 5-HT₃ receptors in both the brain and the gut. Separately, it was shown that escitalopram inhibits 5-HT₃ receptors in [neuronal cancer cells](#). Nevertheless, if serotonin is sufficiently elevated to cause vomiting via 5-HT₃ receptors, even fluoxetine does not inhibit them strongly enough to overcome its suppression of serotonin reuptake and the resulting increase in serotonin activation of the receptors. This was demonstrated [in ferrets](#), in a study where the cancer drug cisplatin caused vomiting and fluoxetine worsened it.

Nevertheless, the fact that 5-HT₃ and 5-HT₄ are the receptors known to be present on the inner mitochondrial membrane and necessary for protection of mitochondrial respiration during hypoxia raises the question of whether fluoxetine or paroxetine enter cells sufficiently to block the receptors on the inner mitochondrial membrane.

A [head-to-head comparison](#) of fluoxetine and escitalopram showed that for both drugs, the intracellular free concentration rapidly becomes equal to the extracellular concentration. However, most of the drug becomes concentrated in intracellular membranes. When the membrane-bound drug is considered, fluoxetine enters cells ten times more effectively than escitalopram, being enriched in membranes 180-fold rather than 18-fold. A [separate study](#) showed that fluoxetine was taken up to a four-fold greater extent than fluvoxamine. In general it appears that SSRIs enter cells very effectively, with fluoxetine>fluvoxamine>escitalopram and where the others fit within that hierarchy being currently unclear.

Since fluoxetine is most effective at entering cells *and* at inhibiting 5-HT₃ receptors, fluoxetine would seem to be particularly likely to interfere with the protective effects of serotonin and melatonin during hypoxia.

However, all SSRIs studied enter cells very effectively and some degree of 5-HT₃ inhibition is shared across at least paroxetine, fluoxetine, and escitalopram.

SSRIs Impact Mitochondria by Activating the Sigma-1 Receptor

Once SSRIs enter cells, they activate the sigma-1 receptor, but with [extreme differences in potency](#).

The [sigma-1 receptor](#) is embedded in segments of the endoplasmic reticulum membrane that lie in close proximity to mitochondria, where it is connected to a binding protein that can release it during stress-induced activation. It causes release of calcium into mitochondria, which activates many processes involved in energy metabolism, and it travels to the plasma membrane, where it can facilitate other forms of cell signaling.

Its natural function is poorly understood. However, the [natural compounds that interact with it](#) include DHEA sulfate, an activator, and pregnenolone, an inhibitor.

Other natural compounds that bind to it include choline, certain fatty substances such as sphingolipids and myristic acid, and the endogenous hallucinogen *N,N*-dimethyltryptamine (DMT), which is also the main ingredient of ayahuasca.

Notably, DMT is synthesized from tryptophan like melatonin and serotonin, and, like serotonin, rapidly broken down by monoamine oxidase. Since monoamine oxidase requires oxygen, DMT concentrations likely rise in hypoxia and activate the sigma-1 receptor.

[In rodents](#), the sigma-1 receptor is responsible for sensitivity to the glucocorticoid receptor during stress induced by restraining the animal, and its absence leads to a depressive phenotype with a chronically overactivated hypothalamic-pituitary-adrenal axis.

However, [rodents lacking this receptor](#) are also less likely to despair when battling humans who are restraining them, and they maintain a more robust preference for sugar than their normal counterparts, which is considered an experiment sign of the ability to enjoy pleasure.

They have higher brain and plasma serotonin and higher plasma histamine, suggesting that activation of the sigma-1 receptor lowers the synthesis or increases the degradation of histamine and serotonin.

The sigma-1 receptor is [necessary for autophagy](#), which at the mitochondrial level makes it important for [mitophagy](#), the clearance of damaged mitochondria, which prevents mitochondrial damage from causing cellular death.

It also has [specific effects](#) on mitochondrial metabolism. It strongly increases complex I activity, which causes a mild increase in reactive oxygen species that elicits the

activation of protective antioxidant pathways. It increases the activity of the other respiratory chain complexes but the impact is much more mild. Its activation of calcium entry into the mitochondria [likely causes](#) increases in the activities of pyruvate, isocitrate, and malate dehydrogenases, which increase the production of NADH that is oxidized by complex I. Mitochondrial metabolism that begins with complex I leads to the synthesis of 40% more ATP than metabolism that begins later in the respiratory chain, so the likely impact of sigma-1 receptor activation is a large increase in the most efficient means of producing ATP in response to energy-demanding stressors.

Activation of the sigma-1 receptor also protects against hypoxia. For example, in [human neurons and immune cells](#), DMT dramatically improved cell survival during oxygen deprivation and prevented the rise in HIF activation, and this was blocked by impairing the sigma-1 receptor.

In [live rats](#) with their cerebral blood supply cut off, DMT maintains the ATP-dependent pumping of ions needed to maintain normal neurological function. The mechanism of the protective effect was not shown in these papers, but it is reminiscent of the effect of melatonin: it *decreases* activation of the hypoxia response while *promoting* cell survival and ATP-dependent functions, suggesting it is somehow *enabling* the paradoxical use of the respiratory chain to make ATP in the absence of oxygen.

Consistent with this, [in neurons from the retina of rats](#), the synthetic opioid pentazocine, which activates sigma-1 receptors, preserved the ability of the respiratory chain to pump protons during deprivation from glucose and oxygen. It also increased complex IV activity and at least one protein involved in transport of substances across the mitochondrial membrane.

Again this is consistent with the effects of serotonin and melatonin. As in our discussion of their impacts, however, there is clearly a missing piece of the puzzle. Complex IV being more active in an in vitro assay does not explain how it is facilitating respiratory chain activity within the cell during exposure to hypoxia, as complex IV requires oxygen to do anything.

[In mice](#), tightly tying up a nerve with a suture activates the sigma-1 receptor, and this increases the expression of the serotonin transporter as well as tryptophan hydroxylase, the rate-limiting enzyme involved in intracellular serotonin synthesis. Thus, the sigma-1 receptor probably protects against hypoxia by bringing serotonin into the cell, synthesizing even more serotonin within the cell, and allow the serotonin to be converted to melatonin inside the mitochondria. The melatonin could then substitute for oxygen as described in the [last installment](#).

I suspect that the sigma-1 receptor also increases the expression or activity of the mitochondrial enzymes that convert serotonin to melatonin, but this has not been studied. It is interesting that methylation is needed to get rid of histamine and to convert serotonin to melatonin. The rise in histamine and serotonin in mice lacking the sigma-1 receptor could reflect impairment in both of these processes.

The order of potency of SSRI activation of sigma-1 receptors is fluvoxamine>sertraline>fluoxetine>escitalopram>citalopram>paroxetine. Fluvoxamine and sertraline are relatively similar in their potency, and citalopram and escitalopram are quite close to one another, whereas fluoxetine is half that of sertraline, citalopram and escitalopram are about half that of fluoxetine, and paroxetine is almost ten times less potent than citalopram and escitalopram. The gap between the most potent (fluvoxamine) and the least potent (paroxetine) is 52-fold. Thus, the differences are extreme, and we can summarize them by saying paroxetine is so trivial as to be irrelevant, while the potent activators are fluvoxamine and sertraline, and the others are relatively weak activators.

If we compare this order of activation to the sporadic and inadequate animal studies cited above suggesting that citalopram and paroxetine may lower serotonin synthesis while sertraline may raise it, we can see this lines up *perfectly* with sertraline being a strong activator of the sigma-1 receptor, citalopram being weak, and paroxetine being trivial.

Thus, it is likely that the *general* effect of SSRIs as a class is to inhibit serotonin synthesis by increasing serotonin activation of extracellular serotonin receptors and thereby stimulating negative feedback mechanisms, while the effects of *specific* SSRIs are determined by the degree to which they activate the sigma-1 receptor. Strong activation will cancel out the impairment of the serotonin receptor by increasing the expression of the receptor and add on top of this an increase in intracellular serotonin synthesis.

A question arises whether *chronic, sustained* activation of the sigma-1 receptor with a drug is good or bad for mitochondria. On the one hand, clearing away damaged mitochondria is an unambiguously good thing. However, could it oppose mitochondrial biogenesis to the point that it decreases mitochondrial density over the long-term? As I covered in my article, [“Is Urolithin A the Ultimate Longevity Supplement?”](#) it is possible that promoting mitophagy may itself lead to subsequent mitochondrial biogenesis. This is supported by studies with the mitophagy activator urolithin A in worms, where on day one they have decreased mitochondria and their ATP is cut in half, but their mitochondrial density recovers by day 8. It also has vague support from human trials, where there is low-quality evidence suggesting increased mitochondrial biogenesis. However, the regulation of these pathways is not understood well enough to predict whether *everything* that promotes mitophagy will *necessarily* promote mitochondrial biogenesis over the long-term, and there are no studies addressing this question with chronic use of pharmacological sigma-1 receptor activators.

Effect of SSRIs on Respiratory Chain Activity

Once SSRIs enter cells they can also directly interact with mitochondria, either by acting on mitochondrial receptors and mitochondrial serotonin transporters, or by entering the mitochondria and impacting mitochondrial enzymes.

A 2023 study in [pig brain mitochondria](#) showed that escitalopram, fluvoxamine, paroxetine, and sertraline were all powerful inhibitors of complex I; complex II + III activity was more moderately inhibited by paroxetine and sertraline and barely

inhibited by escitalopram and fluvoxamine; complex IV was most powerfully inhibited by escitalopram, somewhat less inhibited by sertraline, even less inhibited by paroxetine, and barely inhibited by fluvoxamine. Complexes II and III were not measured on their own, so it isn't clear whether impacts on the complex II + III measurement reflect complex II, complex III, or the CoQ10 that carries electrons between them.

In this head-to-head comparison, the SSRIs are equally bad when it comes to complex I, while escitalopram seems especially bad when it comes to complex IV.

While that study did not include citalopram or fluoxetine, earlier studies by the same group using the same model showed that [citalopram](#) inhibits complex I the most, complex II to a lesser degree, and has no impact on complex IV; and [fluoxetine](#) is a powerful inhibitor of complex I with little to no impact on any other complexes.

Further, fluoxetine [directly inhibits](#) ATP synthase and accumulates in the mitochondrial membrane where it acts as an uncoupler, which means it causes energy to be lost as heat instead of being used to synthesize ATP. These effects have not been studied for other SSRIs.

With all that said, isolated mitochondria unfortunately strip away the complex effects serotonin would be expected to have at the plasma membrane and that SSRIs would have on the sigma-1 receptor and make everything come down to their impacts on mitochondrial membrane receptors and a possible impact on the transport of serotonin into the mitochondria.

We therefore now turn to in vivo animal experiments.

[In rats](#), one study showed that paroxetine treatment increases activity of respiratory chain complexes in certain parts of the brain, but it did not normalize them to citrate synthase activity, which was [also increased](#). Citrate synthase can be used as an imperfect marker of mitochondrial density, so the parallel increase in citrate synthase and respiratory chain enzymes suggests the major impact was an increase in mitochondrial biogenesis. This casts the respiratory chain data in a less favorable light: complex I was increased in most regions, complex II in fewer regions, complex IV in only one region. This suggests complex II and especially complex IV lagged behind the overall stimulus for mitochondrial biogenesis.

The paroxetine was 10 milligrams per kilogram bodyweight by injection. Injection is often used in rodent studies to avoid issues with rodents disliking the taste and refusing intake and to get around variation in the rate of absorption. This is [the equivalent](#) of a human taking 1.6 milligrams per kilogram bodyweight. Paroxetine is generally not given to children and [given to adults](#) in oral doses of 10-62.5 milligrams per day. For a woman weighing 50 kilograms, this would be 0.2-1.25 milligrams per kilogram bodyweight. Due to clearance by the liver when taken orally, injection is probably twice as potent as oral dosing, making the rat dose the equivalent of 3.2 milligrams per kilogram bodyweight for such a woman. Overall, the dose is roughly around the equivalent of someone with below-average bodyweight taking a maximal dose.

In rats, oral administration of 10 or 20 milligrams per kilogram bodyweight sertraline per day for 21 days mitigated the impairment of complex I otherwise caused by the fish poison [rotenone](#), and 10 milligrams per kilogram bodyweight for 14 days protected against the impairment of complexes I, II, and IV by the fungal toxin [3-nitropropionic acid](#). Since rotenone is a specific inhibitor of complex I, the best explanation for sertraline reversing this is its potent activation of the sigma-1 receptor, which increases complex I activity. The protective effect against the fungal toxin was nullified by nitric oxide donors and synergized with antagonists of nitric oxide production, suggesting that sertraline helped reduce nitric oxide accumulation. Various studies have shown that sertraline has [no effect](#), or [decreases](#) nitric oxide, and the decrease in nitric oxide may be mediated by increasing extracellular serotonin's [activation of 5-HT1A receptors](#).

The sertraline doses are the [human equivalent](#) of 1.6-3.2 milligrams per kilogram bodyweight. Humans six years old and older are [generally given](#) 20-200 milligrams per day of sertraline, which is 0.92 to 9.2 milligrams per kilogram bodyweight for a 48-pound 6-year-old female and 0.28-2.8 milligrams per kilogram bodyweight for a 154-pound adult. Thus, the sertraline doses used in rats line up well with therapeutic doses used in humans.

The [one study on fluvoxamine](#) produced results that were inconsistent across doses and brain regions. The general tendency was for succinate dehydrogenase and complex II (two ways of measuring the same thing) to increase or do nothing, but for the other respiratory chain complexes and citrate synthase to decrease or do nothing. However, the middle dose diverged from this pattern in the prefrontal cortex, where it increased citrate synthase and complex I.

The overall pattern of decreased citrate synthase would be consistent with fluvoxamine activating the sigma-1 receptor to promote mitophagy, leading to destruction of damaged mitochondria and decreased mitochondrial density.

The fact that **paroxetine and fluvoxamine appear to have completely opposite effects on mitochondrial density** is consistent with the fact that fluvoxamine is the most potent SSRI activator of the sigma-1 receptor, which promotes mitophagy, whereas paroxetine has essentially no potency against this receptor, but will increase extracellular serotonin and its activation of mitochondrial biogenesis through 5-HT2A and 2B receptors.

Why complex II/succinate dehydrogenase was increased is unclear, but it could be related to this being the only mitochondrial respiratory chain complex whose subunits are encoded entirely by the nuclear genome. For example, the decrease in mitochondrial density could have hurt the rate of mitochondrial protein translation, while complex II could be made entirely with nuclear protein translation, and could therefore be the complex most easy to replace.

This raises a very important question: **should we expect the potent sigma-1 receptor activators, fluvoxamine and sertraline, to lead to a permanent decline in mitochondrial density, or should we expect the long-term effect to be a bounce-back with healthier mitochondria?**

The answer is not clear at all and demands further study.

The fluvoxamine study injected the rats for two weeks with 10, 30, or 60 milligrams per kilogram bodyweight, which is [the equivalent](#) of a human being taking 1.6, 3.2, or 4.9 milligrams per kilogram bodyweight, though injection may make them closer to the equivalents of 3.2, 6.4, or 9.8 milligrams per kilogram bodyweight. [Doses in humans](#) are 25-300 milligrams per day orally for both adults and children. For an 8-year-old female weighing 56 pounds, this would be 1-12 milligrams per kilogram bodyweight, so the doses in rats line up with what children and adolescents might be given.

Of the two in vivo studies with escitalopram, the first showed that, [on its own](#), it has no effect on complex IV or citrate synthase, but hurts activities of complexes I and II and the flow of electrons from complex II to complex III in some brain regions but not others. [The second](#) showed that on its own it had no impact on any of these parameters overall in the brain, but when combined with the fungal toxin 3-nitropropionic acid, it dose-dependently rescued the deficits in complexes I, II, and IV otherwise caused by the toxin.

The second study did not look at separate brain regions; it simply removed the cerebellum and homogenized the rest of the brain. The first study found that the greatest negative effect on complex I was in the cerebellum, with the striatum and hippocampus being next in line, whereas there was little to no effect in the cortex. The cortex makes up [just under half](#) of a rat's brain after removing the cerebellum, whereas the other structures investigated in the first study only represent about 15% of the brain. Thus, the second study may have failed to uncover harms of escitalopram in specific areas of the brain because they were diluted by the overwhelming dominance of the cortex, where it has no effect on its own but appears to protect against at least one mitochondrial toxin.

A separate study on [sertraline protection](#) against 3-nitropropionic acid toxicity suggested this toxin raises nitric oxide levels and sertraline protects against it by blunting this rise. Escitalopram has been shown to [reduce nitric oxide levels](#), which has been shown to mediate [erectile dysfunction](#) and [reduced anxiety](#), while some studies indicate its impact on nitric oxide production in the brain could be up or down [depending on the brain region](#). Since some extracellular serotonin receptors increase nitric oxide and others decrease nitric oxide, and since some activate production of reactive oxygen species that could scavenge nitric oxide, it is likely this protective effect is tissue-specific, brain region-specific, and mediated by serotonin action on these extracellular receptors.

On the other hand, escitalopram may be directly hurting complexes I and II in certain brain regions, and this could be related to a mix of direct negative impacts on mitochondria in combination with biasing serotonin toward extracellular receptors that can activate the hypoxia response and away from protective intracellular roles of serotonin and melatonin on respiratory chain function during the hypoxia response.

Escitalopram stands out as entering cells at a lower rate than fluvoxamine and fluoxetine and being a relatively weak activator of the sigma-1 receptor, though nowhere near as weak as paroxetine.

These studies used 10-20 milligrams per kilogram of escitalopram, which is [the equivalent](#) of a human taking 1.6-3.2 milligrams per kilogram bodyweight, though injection may make it closer to 3.2-6.4 milligrams per kilogram bodyweight. Escitalopram is [usually used](#) in doses of 5-20 milligrams per day in both adults and children. For an 8-year-old female weighing 56 pounds this would be 0.2-0.8 milligrams per kilogram body weight, so these doses of escitalopram are a bit higher than what pretty much anyone would be given, but are within an order of magnitude.

Fluoxetine is the most studied out of any of the SSRIs. The in vivo fluoxetine studies found the following:

- For the [respiratory chain](#) in rats, acutely 10 milligrams per kilogram had no effect but 25 milligrams per kilogram increased complex I activity in the hippocampus but not in other regions measured; it had no effect after 28 days when examined two hours after the last dose; however, after 24 hours after the last dose, the high dose reduced complex II + III activity in the striatum and both doses reduced complex IV activity in the hippocampus, with no effects in other regions. In a different publication but likely the [same study](#), there was no effect on citrate synthase. Overall the increased complex I could reflect acute activation of the sigma-1 receptor, and the emergence of decreased respiratory chain activity after 24 but not 2 hours suggests the cell had become dependent on increased intracellular serotonin synthesis from chronic sigma-1 activation, which then hurt the respiratory chain activity when it was withdrawn.
- In [rat liver](#), 10 milligrams per kilogram for 12 days appeared to accumulate in mitochondrial membranes and act as an uncoupler, which promotes the use of energy to generate heat without making ATP. In general, mild uncoupling under proper conditions can promote weight loss and help generate heat when it is needed but chemically inducing indiscriminate uncoupling can lead to toxicity at high enough doses.
- In [congenitally helpless rats](#), who are selectively bred to exhibit signs of depression, 5 milligrams per kilogram fluoxetine per day for two weeks increased complex IV activity in one brain region and decreased it in another. They considered this an increase or decrease in “metabolism” and the authors did not measure citrate synthase activity or the activity of any other respiratory chain complexes, making it impossible to put the changes into the context of an overall pattern. Further, they measured complex IV activity in over forty brain regions and considered statistical significance to be $P < 0.05$. They did not even include a statistics section so presumably did not make adjustments for making so many comparisons, and you would expect two spurious results at that significance level with over 40 comparisons made. Thus, this study should be disregarded as noise.
- In [male and female rats](#), 5 milligrams per kilogram bodyweight per day fluoxetine was injected into control animals or animals subjected to social isolation for 21 days. Again, complex IV activity was measured without measuring activity of citrate synthase or other respiratory chain complexes.
 - In female rats, complex IV went up in the prefrontal cortex in response to both stress and fluoxetine, but there was no interaction between the two stimuli,

while in the hippocampus complex IV only went up in response to the combination of stress and fluoxetine.

- In male rats, complex IV went down in the prefrontal cortex in response to both stress and fluoxetine but there was no interaction between the two stimuli, while in the hippocampus it went up in response to fluoxetine alone and down in response to stress, while fluoxetine had no ability to rescue the negative effect of stress.
- The authors were interested in the glucocorticoid receptor as mediating this effect, but this research was done before the importance of the sigma-1 receptor was appreciated. The authors later published [another paper](#) that seems derived from the same study implicating fluoxetine-mediated changes in estrogen receptors in the results.
- Most likely the findings reflect interactions between sex hormones or sex hormone-influenced substances, stress, fluoxetine, and sigma-1 receptor signaling, which is necessary for glucocorticoid sensitivity, but the limited measurements of respiratory chain function make it impossible to properly situate the complex IV changes into an overall pattern and the interactions of various hormonal stimuli with sigma-1 receptors are even now still poorly understood.
- [Newborn male rats](#) were given 10 milligrams per kilogram bodyweight fluoxetine for the first 21 days of their lives. Fluoxetine increased citrate synthase activity in both the hypothalamus and the skeletal muscle, consistent with a stimulation of mitochondrial biogenesis. It increased uncoupling in both tissues, consistent with the previous study shown in rat liver. Indeed, fluoxetine lowered bodyweight, consistent with uncoupling favoring the wasting of energy as bodyheat. This study did not investigate effects on specific respiratory chain complexes.
- In [adolescent rats](#) subjected to four weeks of social isolation with or without 21 days of 7.5 milligrams per kilogram bodyweight fluoxetine administered orally through drinking water, social isolation lowered complex II activity and ATP production in brain and heart. Fluoxetine partly mitigated the decrease in complex II activity in the brain and fully mitigated it in the heart, while it fully restored ATP production in the brain and did nothing to protect ATP production in the heart. The activities of citrate synthase and other respiratory chain complexes were not measured so we cannot properly situate the complex II activity into an overall pattern, but counteracting psychological stress suggests fluoxetine is improving the expected sigma-1 receptor activity response to that stress.
- In [male rats](#) injected with 10 milligrams per kilogram body weight per day for 21 days, in mitochondria of the frontal cerebral cortex, fluoxetine had no impact on citrate synthase but decreased complex II activity in “light” synaptic mitochondria and increased complex IV activity in “heavy” synaptic mitochondria and the mitochondrial of neuronal cell bodies.
 - The names for “light” and “heavy” mitochondria at the synapse refer to how they can be separated by density but their functional differences are not well understood.

- Complexes I and III in this study were not measured separately, but were instead measured as a single unit looking at the ability of NADH to reduce cytochrome C, which was unchanged.
- The hypoxia response can sometimes lead to specific suppression of complex II, so activation of this response through extracellular serotonin receptors could explain the effect found in light synaptic mitochondria. The increase in complex IV could perhaps reflect improved phosphorylation status of complex IV mediated by increased intracellular serotonin synthesis leading to increased activation of mitochondria 5-HT₄ receptors. The diversity of responses between different mitochondria within the same neuron emphasizes that the composition of receptors and other proteins can drive very disparate responses to the same SSRI.
- A [second paper](#) likely derived from the same study just described reporting the hippocampus instead of the cerebral cortex found that fluoxetine increased complex II and IV activities in the mitochondria of cell bodies but decreased complex II activity in light synaptic mitochondria and had no effect in heavy synaptic mitochondria. These results partially overlap with those reported for the cerebral cortex but differ somewhat, and probably reflect the same mechanisms but are different due to different distributions of receptors and other proteins between the different brain regions.

Most of these studies used 5-10 milligrams of fluoxetine per kilogram bodyweight per day administered by injection, though two administered it in drinking water, one of which used 18 milligrams per kilogram bodyweight. Injection could double the bioavailability to 10-20 milligrams per bodyweight per day. 5-20 milligrams per kilogram bodyweight is [the equivalent](#) of a human taking 0.8-3.24 milligrams per kilogram bodyweight. The [doses used in humans](#) are 10-60 milligrams per day for ages 8-17 and as high as 80 milligrams per day in adults. For an 8-year-old, 56-pound female, these are 0.4-2.4 milligrams per kilogram bodyweight, and for a 154-pound adult they are 0.1-1.1 milligrams per kilogram bodyweight. Thus, these reflect the higher end of what might be given to a healthy-weight adult and overlap quite well with what might be given to a child.

How Does Prozac Act As a Performance-Enhancing Drug?

We are now equipped to answer a question raised in the [first article](#) in this series, how Prozac (fluoxetine) acts as a performance-enhancing drug.

In that article, we reviewed a [2019 paper](#) showed that, in mice, fluoxetine is a performance-enhancing drug.

The answer is that it acted as a mitochondrial-enhancing drug.

Fluoxetine augmented the increase in skeletal muscle citrate synthase that occurred in response to exercise, but it didn't increase citrate synthase in sedentary mice.

Exercise and fluoxetine seemed to synergistically increase complex IV activity, but variability prevented the statistics from showing the interaction robustly and statistical significance was only shown between the exercised mice with and without fluoxetine.

No other specific respiratory chain complexes were measured, making it difficult to interpret complex IV in the context of an overall pattern.

The likely effect here was that fluoxetine activated sigma-1 receptors in muscle, increasing intracellular serotonin synthesis, which acted on mitochondrial serotonin receptors and enabled mitochondrial melatonin synthesis. This could have resulted in improved ATP production, which is needed for both muscle protein synthesis and mitochondrial biogenesis. This could also have preserved respiratory chain function during the relative hypoxia of exercise, improving performance.

With that said, the studies collectively show fluoxetine could have many different effects on mitochondria. These include promoting mitochondrial biogenesis, enhancing mitochondrial function by activating the sigma-1 receptor, activating or inhibiting the hypoxia response, increasing intracellular serotonin synthesis to protect mitochondrial respiration, serving as a mitochondrial uncoupler, and directly inhibiting ATP synthase.

Fluoxetine seems to be more often protective of mitochondria than not, but with conflicting effects in different tissues, brain regions, and even in different mitochondria within a single cell.

Not All SSRIs Are the Same

Some of the differences in SSRIs that stand out are as follows:

- Sertraline and fluvoxamine are powerful activators of the sigma-1 receptor while the other SSRIs are not. This is acutely beneficial to mitochondrial function, and critical to protection of respiration during hypoxia, though there is an outstanding question of whether long-term use chronically promoting mitophagy would be good by eliminating damaged mitochondria or bad by excessively reducing mitochondrial density.
- While all SSRIs have the capacity to promote mitochondrial biogenesis through extracellular serotonin signaling, sertraline and fluvoxamine are uniquely potent in their ability to balance this with sigma-1-mediated mitophagy.
- While all SSRIs can promote the cellular hypoxia response by increasing extracellular serotonin and can inhibit the transport of serotonin into the cell where it would be needed to preserve cellular respiration during the hypoxia response, it is at least the case that fluoxetine and escitalopram, and to a lesser extent paroxetine, may stand out in their ability to inhibit that intracellular preservation of respiration by inhibiting 5-HT₃ and 5-HT₄ receptors. Whether other SSRIs inhibit these receptors is less clear. However, it may be more significant that the well-characterized differences in sigma-1 activation preserve respiration in the face of hypoxia, and this would put fluvoxamine and sertraline in the net protective camp, put fluoxetine in the middle, and put citalopram, escitalopram and paroxetine in the harmful anti-respiration bucket.
- While all SSRIs have the capacity to decrease nitric oxide synthesis, which can cause sexual dysfunction but can also protect mitochondria from nitric oxide-based toxins, they all impair respiration when added to isolated mitochondria, and escitalopram seems to perform the worst in such studies.

- Fluoxetine stands alone as having been shown to inhibit ATP synthase and act as a mitochondrial uncoupler, but whether other SSRIs do this hasn't been studied.
- In vivo studies in rodents suggest that paroxetine in net increases mitochondrial biogenesis whereas fluvoxamine in net increases mitophagy and escitalopram in net has no effect on markers of mitochondrial density. Chronic use of fluoxetine has no effect on markers of mitochondrial density in adult rats, but withdrawal from fluoxetine hurts them, suggesting mitochondria become dependent on it for a normal signal stimulating biogenesis. However, fluoxetine does increase mitochondrial biogenesis in newborns and in rats that follow an exercise program.

So What Does This Say About SSRI Withdrawal?

Armed with an understanding of SSRIs as mitochondrial drugs, we can now look at how this shapes the side effects of using them and withdrawing from them.

For this, stay tuned for the next installment.

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